WORKED SOLUTION

STEP III 2002, Question 5

Pure • Factorising Quartic Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q5

The key idea

A quadratic is a constant multiple of a perfect square exactly when its discriminant vanishes. That is the whole of the first part, and it is also the engine of the rest: if we can choose t so that a square of a quadratic minus the quartic is a multiple of a perfect square, then the quartic is a *difference of two squares*, and it factorises at once. This is Ferrari's classical method for the quartic, and it trades the quartic for a cubic in t.

Solution

The quadratic condition

Suppose $p \neq 0$. Completing the square,

$$px^2 + qx + r = p\left(x + \frac{q}{2p}\right)^2 + r - \frac{q^2}{4p}.$$

This has the form $p(x+h)^2$ for some h exactly when the constant $r - \frac{q^2}{4p}$ is zero, that is, when

$$q^2 = 4pr,$$

and then $h = \frac{q}{2p}$. (Conversely, expanding $p(x+h)^2 = px^2 + 2phx + ph^2$ gives $q^2 = 4p^2h^2 = 4pr$, so the condition is necessary as well as sufficient.) If $p = 0$, the only possibility is $q = r = 0$; the condition $q^2 = 4pr$ still forces $q = 0$, but we also need $r = 0$.

The equation for t

Expanding,

$$(x^2 + \tfrac{1}{2}bx + t)^2 = x^4 + bx^3 + (\tfrac{1}{4}b^2 + 2t)x^2 + btx + t^2.$$

The choice of $\frac{1}{2}b$ is exactly what makes the x^3 terms cancel (and the x^4 terms cancel too) when we subtract the quartic:

$$(x^2 + \tfrac{1}{2}bx + t)^2 - (x^4 + bx^3 + cx^2 + dx + e) = (\tfrac{1}{4}b^2 + 2t - c)x^2 + (bt - d)x + (t^2 - e).$$

This is a quadratic with $p = \frac{1}{4}b^2 + 2t - c$, $q = bt - d$ and $r = t^2 - e$, and we want it to equal $k(x+h)^2$, a multiple (namely $k = p$) of a perfect square. By the first part, t must satisfy

$$(bt - d)^2 = 4(\tfrac{1}{4}b^2 + 2t - c)(t^2 - e).$$

This is a cubic equation in t (the t^3 term on the right has coefficient 8, while the left is only quadratic in t), so it always has at least one real root.

Factorising the quartic

Here $b = 6$, $c = 9$, $d = -2$, $e = -7$, so

$$\tfrac{1}{4}b^2 + 2t - c = 2t, \qquad bt - d = 6t + 2, \qquad t^2 - e = t^2 + 7.$$

The equation for t becomes

$$\begin{aligned}(6t+2)^2 &= 8t(t^2+7)\\ 36t^2 + 24t + 4 &= 8t^3 + 56t\\ 0 &= 2t^3 - 9t^2 + 8t - 1.\end{aligned}$$

Ask yourself How do we find a root of this cubic quickly?

Try small values first: the coefficients add up to $2 - 9 + 8 - 1 = 0$, so $t = 1$ is a root. Any root will do, because each one gives a valid factorisation; the simplest root gives the simplest one.

With $t = 1$ the difference is

$$2x^2 + 8x + 8 = 2(x+2)^2,$$

so $k = 2$ and $h = 2$, and indeed $8^2 = 4 \cdot 2 \cdot 8$. Hence

$$(x^2 + 3x + 1)^2 - (x^4 + 6x^3 + 9x^2 - 2x - 7) = 2(x+2)^2,$$

that is,

$$x^4 + 6x^3 + 9x^2 - 2x - 7 = (x^2 + 3x + 1)^2 - (\sqrt{2}\,(x+2))^2.$$

This is a difference of two squares, so

$$\begin{aligned}x^4 + 6x^3 + 9x^2 - 2x - 7 &= (x^2 + 3x + 1 - \sqrt{2}(x+2))(x^2 + 3x + 1 + \sqrt{2}(x+2))\\ &= (x^2 + (3 - \sqrt{2})x + 1 - 2\sqrt{2})(x^2 + (3 + \sqrt{2})x + 1 + 2\sqrt{2}).\end{aligned}$$

Because $k = 2 > 0$, its square root is real, so these are two quadratic factors with real coefficients, as required.

The other two roots of the cubic, $t = \frac{1}{4}(7 \pm \sqrt{41})$ from $2t^2 - 7t + 1 = 0$, also have $k = 2t > 0$, and they give the other two ways of pairing up the four (real) roots of the quartic into real quadratic factors. Their coefficients are much messier, which is why $t = 1$ is the one to use.

Checks

- Multiply out: the product of the two factors has x^3 coefficient $(3-\sqrt{2})+(3+\sqrt{2}) = 6$ and constant term $(1-2\sqrt{2})(1+2\sqrt{2}) = 1-8 = -7$, both correct.
- At $x = 1$ the quartic is $1+6+9-2-7 = 7$, and the factorisation gives $(5-\sqrt{2}-2\sqrt{2})(5+\sqrt{2}+2\sqrt{2}) = 25-18 = 7$.
- The discriminants of the factors are $7+2\sqrt{2} \approx 9.83$ and $7-2\sqrt{2} \approx 4.17$, both positive, so the quartic has four real roots, consistent with all three roots of the cubic giving real factorisations. We also checked the factorisation numerically at 200 random points.

Common slips

- Getting the x^2 coefficient of $(x^2+\frac{1}{2}bx+t)^2$ wrong: it is $\frac{1}{4}b^2+2t$, since the cross term $2\cdot x^2\cdot t$ contributes as well as $(\frac{1}{2}bx)^2$.
- Writing the discriminant condition as $q^2 = pr$ or forgetting that k is the leading coefficient $\frac{1}{4}b^2+2t-c$, which must be positive for a real factorisation.

Beyond STEP

The cubic in t is called the *resolvent cubic*, and this is Ferrari's sixteenth-century method: every quartic reduces to a cubic, which Cardano's formula then solves. The deeper reason it works, and the reason there is no such formula for general quintics, belongs to Galois theory, which studies how the roots of a polynomial can be permuted. For a first look at the algebra of fields and polynomials, try Allenby, *Rings, Fields and Groups*.

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